Outlook

RE: Follow up on BRF measurements

From Anderson, Nikolaus J - (nja) <nanderson@optics.arizona.edu>
Date Thu 2/19/2026 12:08 PM
To Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>; Nicolas, Pierre Raphael - (pierrenicolas) <pierrenicolas@arizona.edu>; Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Cc Kim, Daewook - (duey) <dkim@optics.arizona.edu>; Mavko, Matthew Edward - (mmavko) <mmavko@arizona.edu>

Heejoo,

My recommendation is that we try to make a plan to measure samples outside next week, perhaps Thursday pending better weather info.

I feel confident we'll be able to get some reasonable reflectance results outdoors with sample sizes 20mm or larger, which I think you already have.

For full BRF measurements in our Blacklab, my understanding is that we will be limiting that work to one or two coatings which will be the most likely candidates for implementation based on the outdoor spectrometer results. For those sample sizes, let me look through some of our old mounting methods to see what might work best for smaller samples.

We'll plan to circle back early next week and try to plan on a meeting time for outdoor reflectance collection in a good weather window.

Thanks,

Nik

From: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>
Sent: Wednesday, February 18, 2026 11:33 PM
To: Anderson, Nikolaus J - (nja) <nanderson@optics.arizona.edu>; Nicolas, Pierre Raphael - (pierrenicolas) <pierrenicolas@arizona.edu>; Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Cc: Kim, Daewook - (duey) <dkim@optics.arizona.edu>; Mavko, Matthew Edward - (mmavko) <mmavko@arizona.edu>
Subject: Re: Follow up on BRF measurements

Thanks for the elaboration. I'm learning a lot from your emails every time.

We are going to fabricate more samples as shown below.
We might have a chance to ask for the specific sample size to fit your system and make implementation easier.
Do you have any recommendations for the sample size?

- Sample list

Aeroglaze Z307 on Invar and aluminum (the aluminum primer is delayed till march)

Pioneer optical black anodize on aluminum.

Black electroless nickel on Invar

Titanium Anodize on titanium

Best
Heejoo

From: Anderson, Nikolaus J - (nja) <nanderson@optics.arizona.edu>
Sent: Wednesday, February 18, 2026 4:13 PM
To: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>; Nicolas, Pierre Raphael - (pierrenicolas) <pierrenicolas@arizona.edu>; Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Cc: Kim, Daewook - (duey) <dkim@optics.arizona.edu>; Mavko, Matthew Edward - (mmavko) <mmavko@arizona.edu>
Subject: RE: Follow up on BRF measurements

Heejoo,

Sorry, I just kind of blasted through an old excel file and threw this plot together.

Y axis is absolute reflectance, so a value of 1 would be 100% reflectance. (X axis is wavelength in nm.) This older field spectrometer data has issues at the spectral changeovers at 1000nm and ~1800nm, which show up as vertical anomalies. The data in atmospheric absorption regions should essentially be ignored.

We use the brand of field spectrometers known as ASD. So… ASD data has known low light limitations, so trusting an absolute value for a very low reflectance target is not recommended. However, as you can see, you can get a very good sense of the spectral reflectance shape. The absolute value of a low reflectance target should be further studied with a more sensitive instrument (which is what we hope to get out of our Blacklab measurements).

Thanks,

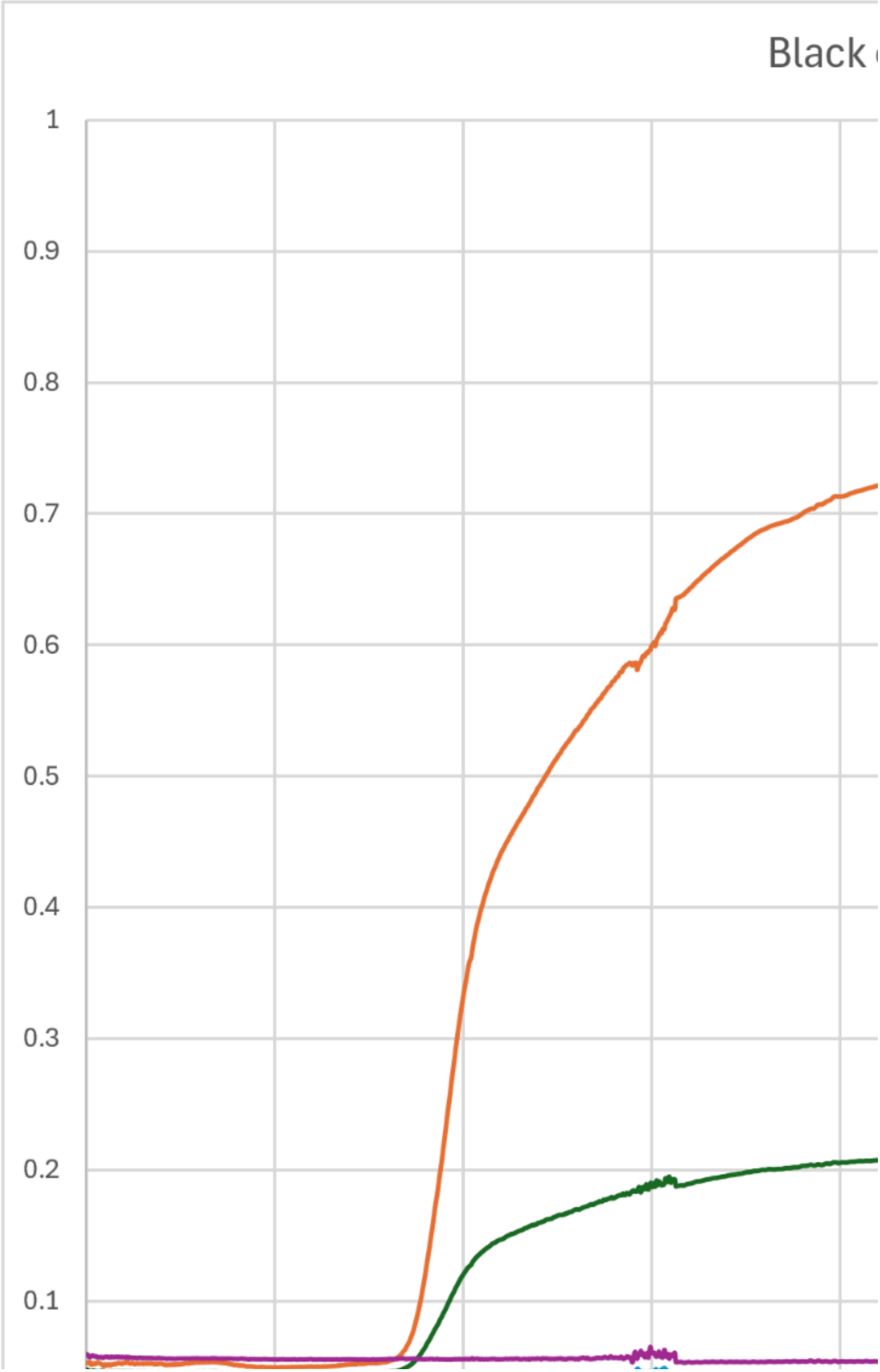
Nik

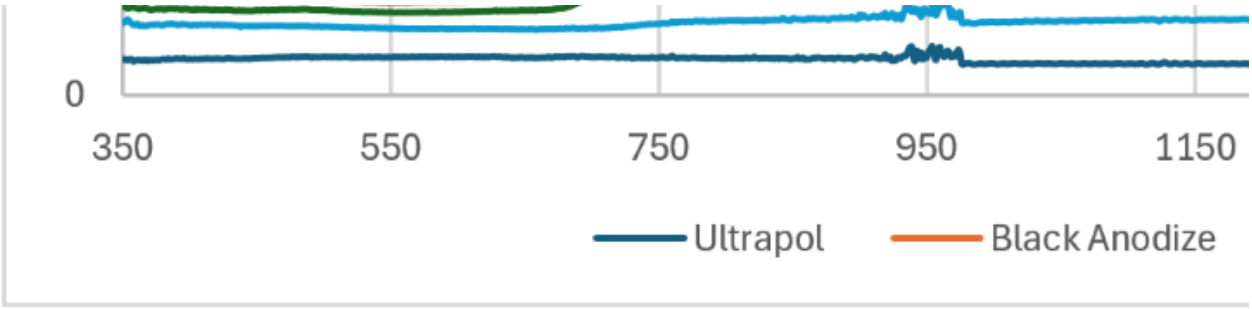
From: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>
Sent: Wednesday, February 18, 2026 3:44 PM
To: Anderson, Nikolaus J - (nja) <nanderson@optics.arizona.edu>; Nicolas, Pierre Raphael - (pierrenicolas) <pierrenicolas@arizona.edu>; Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Cc: Kim, Daewook - (duey) <dkim@optics.arizona.edu>; Mavko, Matthew Edward - (mmavko) <mmavko@arizona.edu>
Subject: Re: Follow up on BRF measurements

Hello Nik

The shared data is really helpful, and the format matches exactly what we expect.
One quick question. What is the unit of the y-axis? Is it normalized to input power?

Many thanks,
Heejoo


Black
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1



Best
Heejoo

From: Anderson, Nikolaus J - (nja) <nanderson@optics.arizona.edu>
Sent: Wednesday, February 18, 2026 3:22 PM
To: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>; Nicolas, Pierre Raphael - (pierrenicolas) <pierrenicolas@arizona.edu>; Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Cc: Kim, Daewook - (duey) <dkim@optics.arizona.edu>; Mavko, Matthew Edward - (mmavko) <mmavko@arizona.edu>
Subject: RE: Follow up on BRF measurements

Somehow in the research into Pierre and Jacob's contact info I forgot to include Matt!

Apologies, and please use this email chain to reply all…

From: Anderson, Nikolaus J - (nja)
Sent: Wednesday, February 18, 2026 10:36 AM
To: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>; Nicolas, Pierre Raphael - (pierrenicolas) <pierrenicolas@arizona.edu>; Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Cc: Kim, Daewook - (duey) <dkim@optics.arizona.edu>
Subject: Follow up on BRF measurements

Good morning,

Sorry for the delay, I had started this email yesterday but never found time to finish.

I hope I've properly found and included Pierre and Jacob on here!

I've attached a few examples of processed output files from lab BRF measurements, and also some results from prior field spectrometer measurements of black coatings on aluminum. The files that start with the date are the multi-wavelength files and are the most likely you'll want to use in the future. The single wavelength files are more specific for compatibility with our solar zenith code, so I just included a couple of those as an FYI.

I have thought things through further, and had some discussions with Matt, and I do think there are some ways we can improve the signal levels looking at black samples. Specifically, we do have a high-quality amplifier with decadal gain settings through ~ 10 orders of magnitude with the VNIR instrument. Typically, we do not set different gains for a sample vs. a reference in order to minimize added uncertainty (and typically they are similar reflectance anyway). But for black samples, this may be worth the slight added uncertainty in order to increase the gain and get a bit more signal out of the radiometer. Anyhow, something to experiment with when we get to that point…

Thanks again, we're excited about this collaboration!

Nik